Global Electric Vehicle Market Analysis

Performance, Pricing, Battery Technology and Sales

SmartBridge Skill Wallet Project: Data Analytics with Tableau

Tools: Python, SQL (SQLite), Tableau Public, Flask

Student name	Pragya Jain
GitHub repository	https://github.com/jainpragya2411-hub/EVM-project.git
Tableau dashboard	https://public.tableau.com/views/Electric_Vehicle_Market_Insights_Sam_17932780750580/ElectricVehicleMarketPerformanceInsights?:language=en-US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link
Tableau story	https://public.tableau.com/views/Electric_Vehicle_Market_Insights_Final_17932783119840/SalesGeography?:language=en-GB&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link
Demo video	

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1. Problem understanding

Electric vehicles are becoming a mainstream product, but manufacturers, policy makers and buyers still struggle to compare the market. Models vary widely in price, range, battery chemistry, charging speed and sales. Without a clear view it is hard to know which vehicle types sell best, what drives price, and which battery technologies last longest.

Objectives

- Measure the size and structure of the EV market by category, manufacturer, region and country.
- Understand how price relates to range, battery capacity and motor power.
- Compare battery technologies (NMC, LFP, NCA, Sodium-Ion, Solid-State) on durability and charging.
- Identify high-value vehicles and the environmental impact (CO_2 saved).
- Deliver the findings as an interactive Tableau dashboard and story embedded in a Flask web app.

Business questions

- Which categories and manufacturers lead in annual sales?
- How strongly do battery capacity and range drive price?
- Which battery type has the best cycle life and lowest degradation?
- How do market regions and manufacturing countries compare?
- Which vehicles give the most range per dollar?
- How much CO_2 do EVs save, by category?

2. Dataset

The dataset **ev_market_test.csv** was provided by Skill Wallet: **100,000 rows and 37 columns**. Each row is one EV model or variant (vehicle_id is unique; no duplicate records). Model years run from 2018 to 2026, with about 11,000 models per year.

Group	Columns
Identity	vehicle_id, model_name, manufacturer (200), model_year, manufacturing_country (10), market_region (7), category (15)
Battery	battery_capacity_kwh, battery_type (5), battery_cycle_life, battery_degradation_pct_5yr
Performance	range_km, efficiency_wh_per_km, motor_power_kw, torque_nm, num_motors, drivetrain, top_speed_kmph, acceleration_0_100_sec
Charging	max_dc_charging_kw, charging_dc_fast_time_min, charging_ac_time_hr, num_charging_ports
Body and features	num_seats, weight_kg, cargo_capacity_liters, infotainment_screen_inch, autonomous_level, safety_rating_stars
Commercial	price_usd, insurance_cost_usd_yearly, warranty_years, resale_value_pct_3yr, annual_sales_units, customer_rating
Impact and use	co2_saved_tons_per_year, recommended_use_case (7)

Data quality

Column	Missing values
resale_value_pct_3yr	3,057
battery_cycle_life	2,028
customer_rating	1,931
infotainment_screen_inch	1,506
annual_sales_units	989

- All other columns are complete. Missing values are under 3.1% per column and are left blank, so averages ignore them.
- Outliers exist by design (buses and trucks have very high efficiency values, seats up to 65, price up to about $410k), so charts compare vehicles within their category.
- The data is synthetic: models per year, regional totals and average range by battery type are almost uniform, so the analysis focuses on fields with real variation.

3. Data stored in a database

The CSV was loaded into a relational database table named **ev_market** (100,000 rows, 37 columns) using Python and pandas. The database is a **SQLite** file (`ev_market.db`). The repository also contains a MySQL version of the loader and queries (`scripts/load_to_db.py`, `sql/queries.sql`).

```
df = pd.read_csv("ev_market_test.csv")
df.to_sql("ev_market", con, if_exists="replace", index=False)
# Stored 100,000 rows in table ev_market (database file: ev_market.db)
```

4. SQL operations

Eight SQL queries were run on the table. They cover counting, aggregation, grouping, sorting, calculated columns and a view that prepares data for Tableau. Screenshots of the output follow.

```
-- Top manufacturers
SELECT manufacturer, CAST(SUM(annual_sales_units) AS INTEGER) AS total_sales
FROM ev_market GROUP BY manufacturer ORDER BY total_sales DESC LIMIT 10;

-- Battery technology comparison
SELECT battery_type, ROUND(AVG(range_km),1) AS avg_range_km,
       ROUND(AVG(charging_dc_fast_time_min),1) AS avg_fast_charge_min,
       ROUND(AVG(battery_cycle_life),0) AS avg_cycle_life,
       ROUND(AVG(battery_degradation_pct_5yr),2) AS avg_degradation_pct
FROM ev_market GROUP BY battery_type ORDER BY avg_cycle_life DESC;

-- Clean view for Tableau
CREATE VIEW vw_ev_clean AS
SELECT *, ROUND(range_km*1.0/price_usd*1000,2) AS range_per_1000usd,
  CASE WHEN price_usd < 20000 THEN 'Budget' WHEN price_usd < 60000 THEN 'Mid-range'
       WHEN price_usd < 120000 THEN 'Premium' ELSE 'Luxury' END AS price_segment
FROM ev_market WHERE price_usd > 0 AND range_km > 0;
```

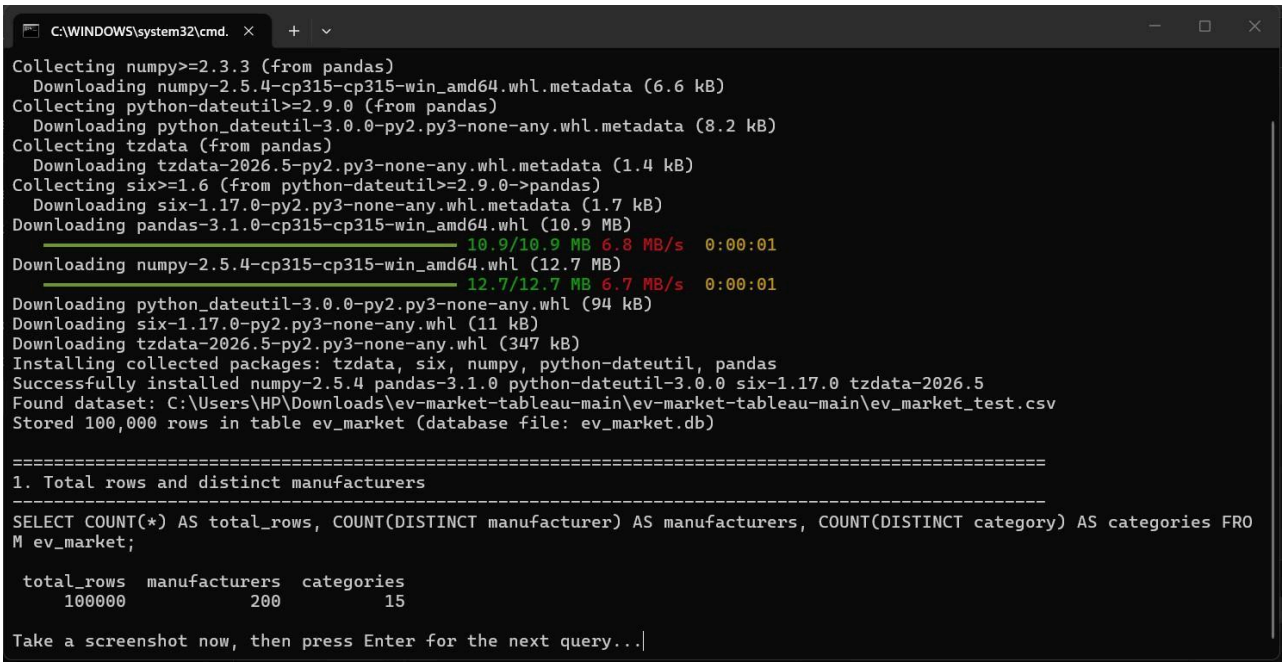


Figure 1: 100,000 rows stored in table ev_market; query 1 shows 100,000 rows, 200 manufacturers, 15 categories.

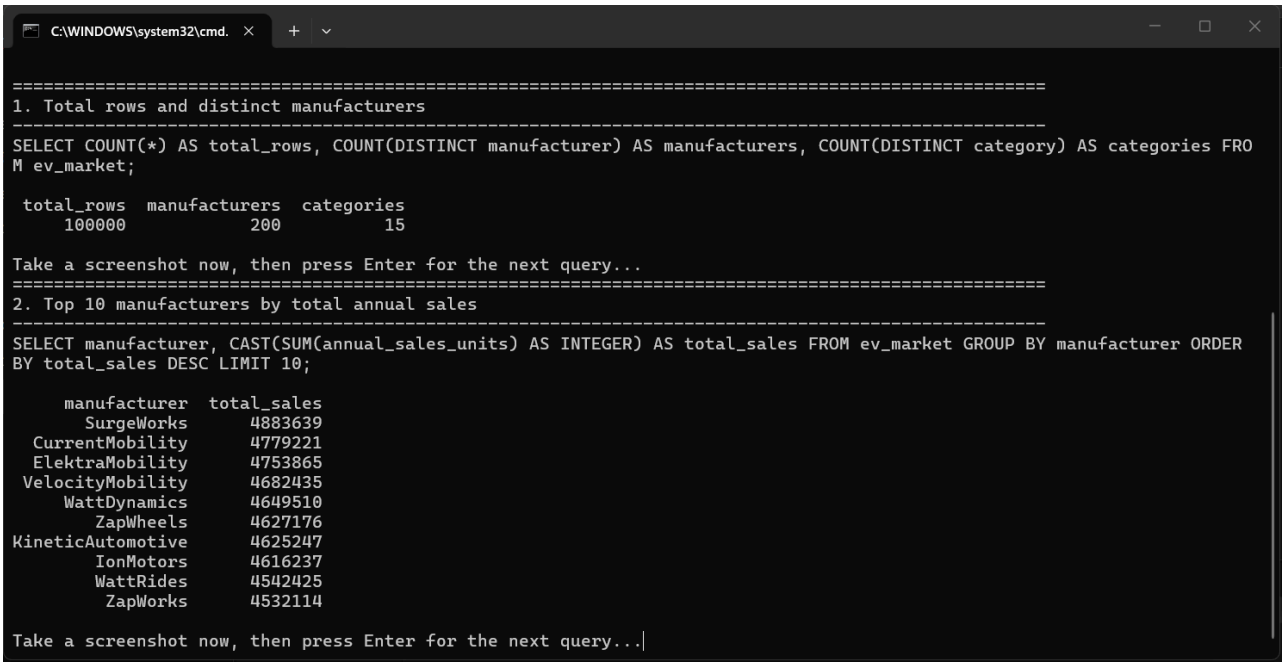


Figure 2: Query 2, top 10 manufacturers by total annual sales.

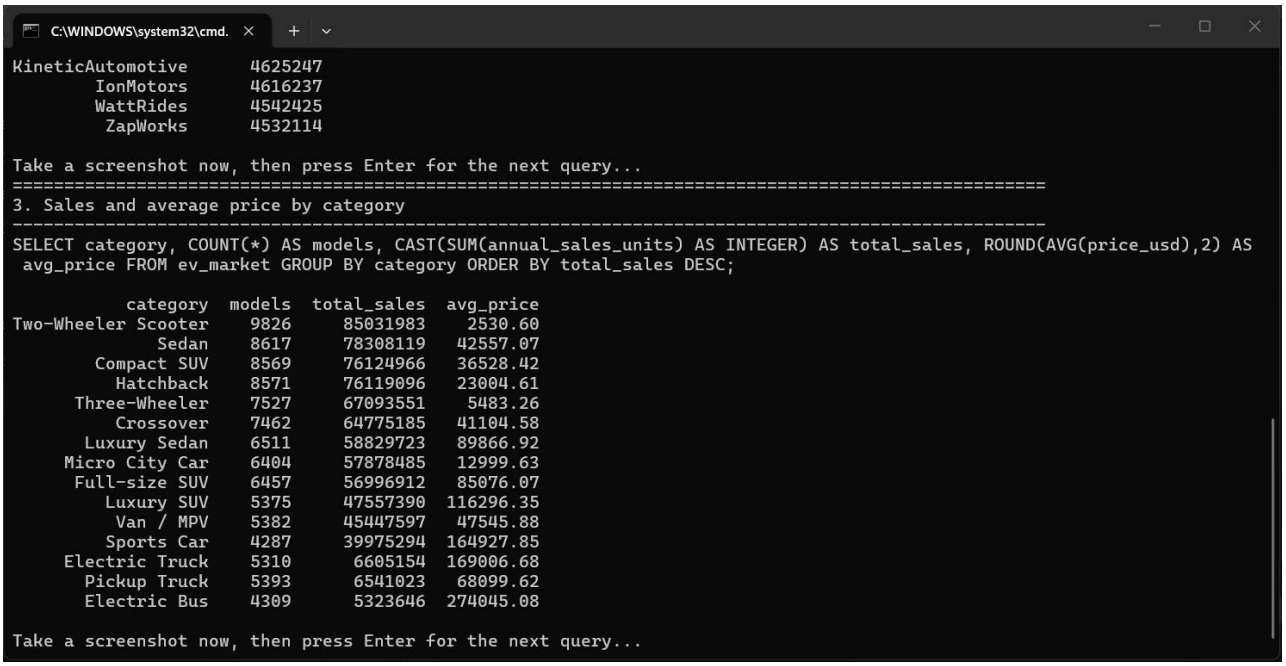


Figure 3: Query 3, sales and average price by category.

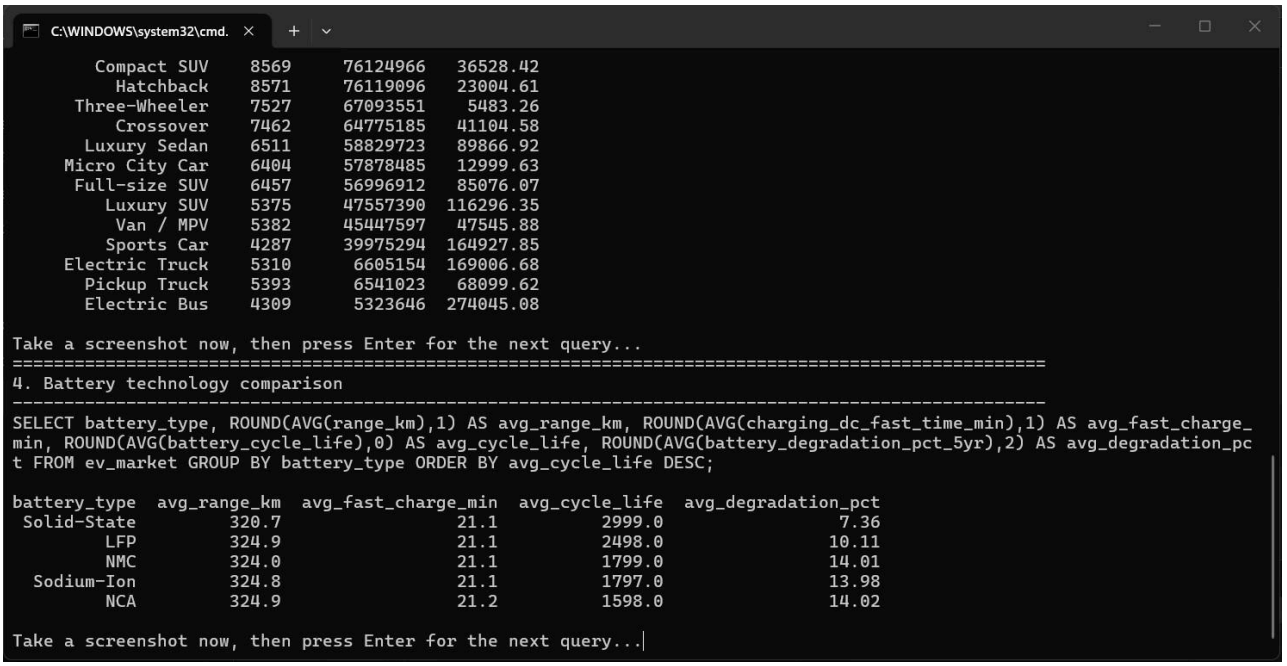


Figure 4: Query 4, battery technology comparison.

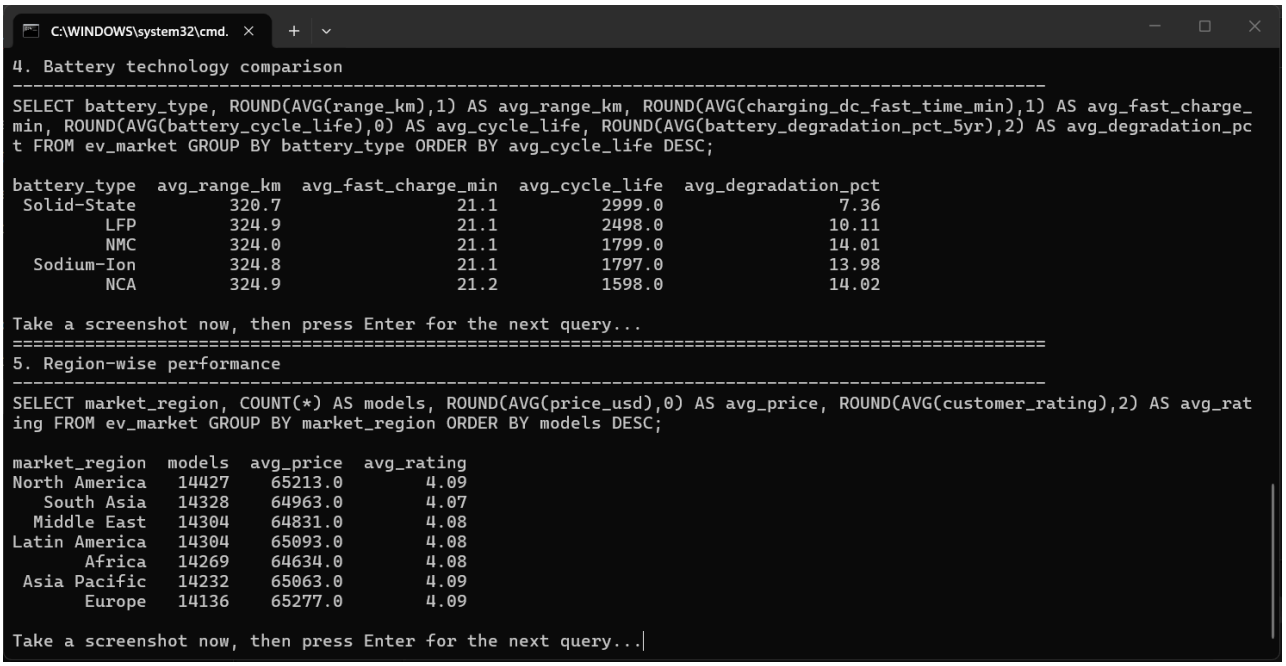


Figure 5: Query 5, region-wise performance.

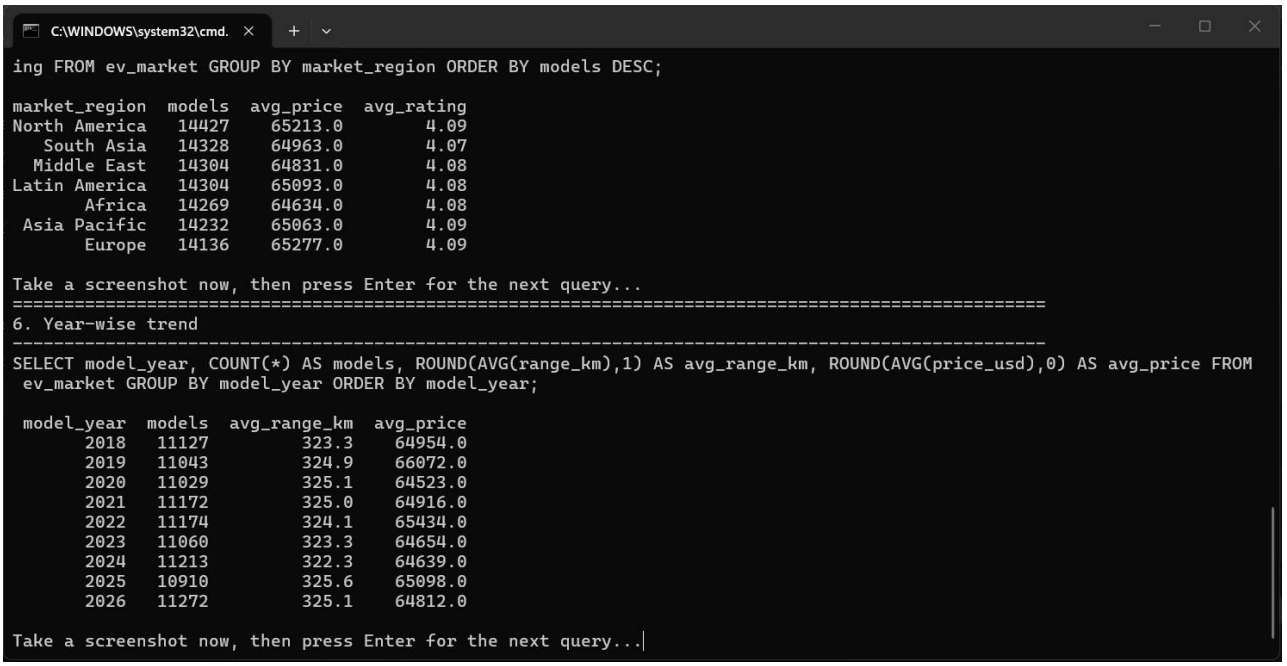


Figure 6: Query 6, year-wise trend.

5. Tableau connection and data preparation

The dataset (the same 100,000 rows that were stored in the database) was connected to Tableau Public as a text file. Tableau Public connects to files rather than database servers.

Data preparation

- **Missing values:** left blank for averages (Tableau ignores nulls); each affected column is under 3.1% missing.
- **Data types:** model_year kept numeric; manufacturing_country given a geographic role for the map.
- **High cardinality:** 200 manufacturers handled with a Top N filter.
- **Derived fields:** Price Segment, Range Class and Fast Charge Rating created as calculated fields.
- **Outliers:** buses, trucks and scooters compared within their own category rather than removed.

Calculated fields

```
Range per 1000 USD = [Range Km] / [Price Usd] * 1000
Price Segment      = IF [Price Usd] < 20000 THEN "Budget" ELSEIF [Price Usd] < 60000 THEN "Mid-range"
                     ELSEIF [Price Usd] < 120000 THEN "Premium" ELSE "Luxury" END
Range Class        = IF [Range Km] >= 450 THEN "Long" ELSEIF [Range Km] >= 250 THEN "Medium" ELSE "Short" END
Fast Charge Rating = IF [Charging Dc Fast Time Min] <= 15 THEN "Ultra-fast"
                     ELSEIF [Charging Dc Fast Time Min] <= 25 THEN "Fast" ELSE "Standard" END
Revenue (Est.)     = [Price Usd] * [Annual Sales Units]
Sales Share        = SUM([Annual Sales Units]) / TOTAL(SUM([Annual Sales Units]))
```

Filters (6)

Model Year (range slider), Market Region, Category, Battery Type, Price Segment and Manufacturer. Each filter is applied to all worksheets that use the data source and is shown on the dashboard.

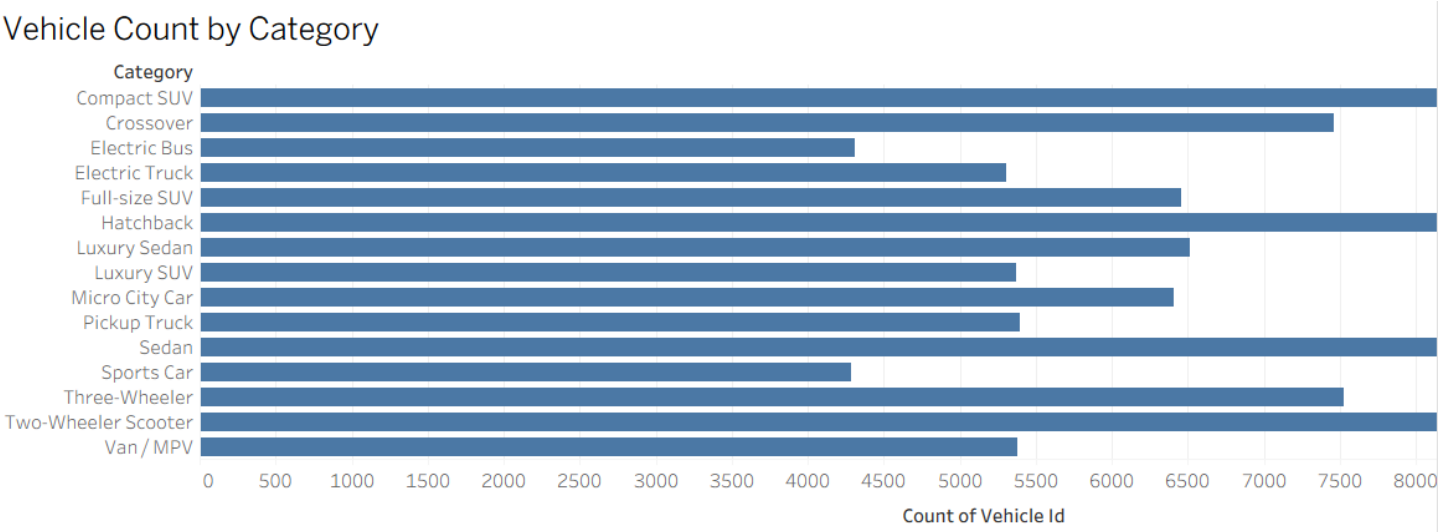
6. Visualizations

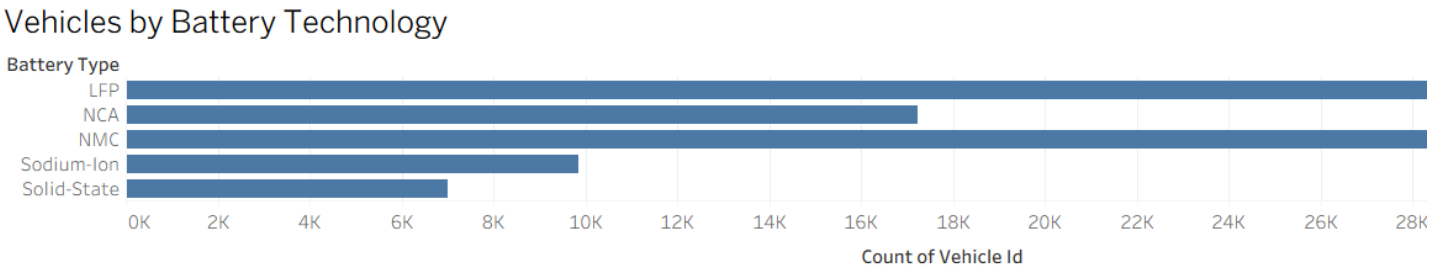
The workbook contains 12 visualizations:

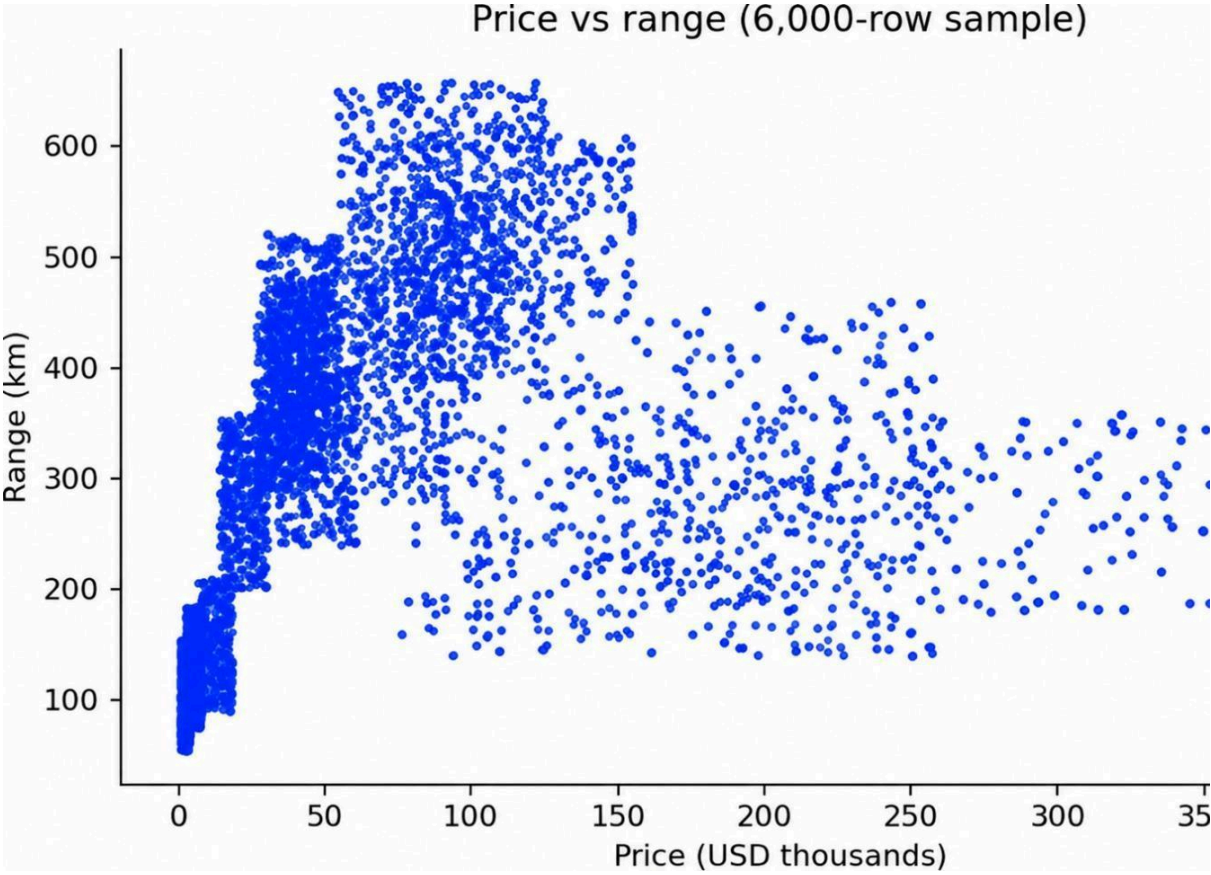
#	Visualization	Type	Question answered
1	Sales by category	Bar	Which vehicle types sell most?
2	Sales by year	Line	Is the market growing?
3	Top manufacturers	Bar + filter	Who leads?
4	Country map	Filled map	Where are models built?
5	Average price by region	Bar	How do regions compare?
6	Battery share	Pie	Which chemistry is most common?
7	Battery durability	Bar	Which battery lasts longest?
8	Price vs range	Scatter	Does more money buy more range?
9	Drivetrain heat map	Heat map	Where do FWD/RWD/AWD concentrate?
10	Recommended use case	Treemap	What are EVs used for?
11	CO2 saved	Stacked bar	Where is the climate impact?
12	Customer rating	Histogram	How satisfied are owners?

Sample visual outputs

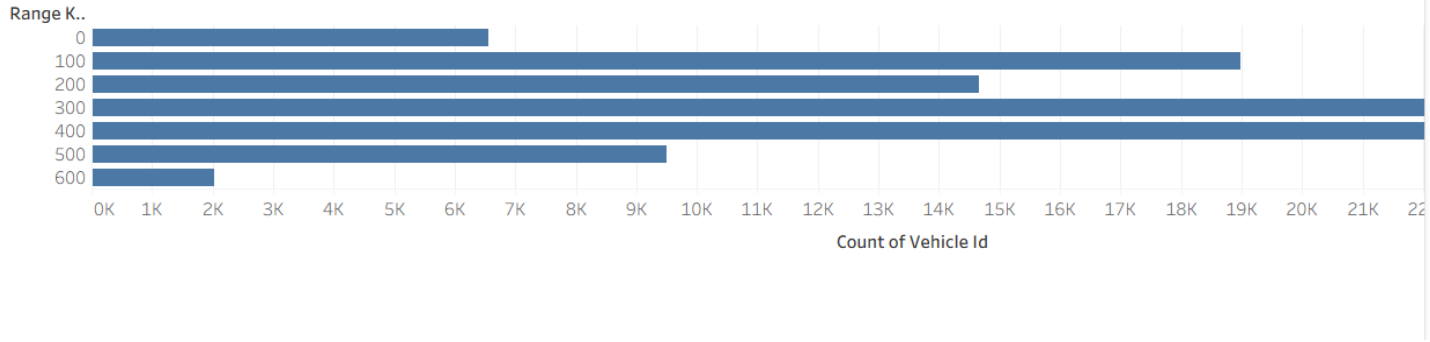
Charts computed from the dataset to show the main patterns (the interactive versions are in the Tableau dashboard and story).







Vehicle Range Analysis.



7. Dashboards

The dashboards combine the worksheets and share the six filters. Dashboard size is set to **Automatic** so the layout adjusts to the browser window, and Phone layouts were available through Device Preview. Clicking a bar or region filters the other charts on the dashboard.

Published dashboard:
https://public.tableau.com/views/Electric_Vehicle_Market_Insights_Sam_17932780750580/ElectricVehicleMarketPerformanceInsights?:language=en-US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

8. Story

The story has six scenes, each with a caption:

Scene	Caption	Message
1	Market snapshot: 100,000 EV models	Size and scope of the market.
2	Who sells the most	Scooters, sedans and compact SUVs lead; trucks, pickups and buses sell far fewer units.
3	Price vs Range	Battery capacity drives price much more than range does.
4	Battery technology	Solid-State and LFP last longest with the lowest degradation.
5	Regions and countries	Regional totals are balanced; no single region dominates.
6	Recommendations	Focus on high-volume categories and durable battery chemistries.

Published story:
https://public.tableau.com/views/Electric_Vehicle_Market_Insights_Final_17932783119840/SalesGeography?:language=en-GB&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

9. Web integration with Flask

The Tableau dashboard and story are embedded in a Flask web application using the Tableau Embedding API. The app has four pages (Home, Dashboard, Story, About), a responsive layout and light and dark mode. The Tableau links are read from a configuration file, so no code changes are needed to update them.

```
@app.route("/dashboard")
def dashboard():
    return render_template("dashboard.html", viz_url=DASHBOARD_URL)
```

```
<script type="module" src="https://public.tableau.com/javascripts/api/tableau.embedding.3.latest.min.js"></script>
<tableau-viz src="{{ viz_url }}" toolbar="bottom" hide-tabs></tableau-viz>
```

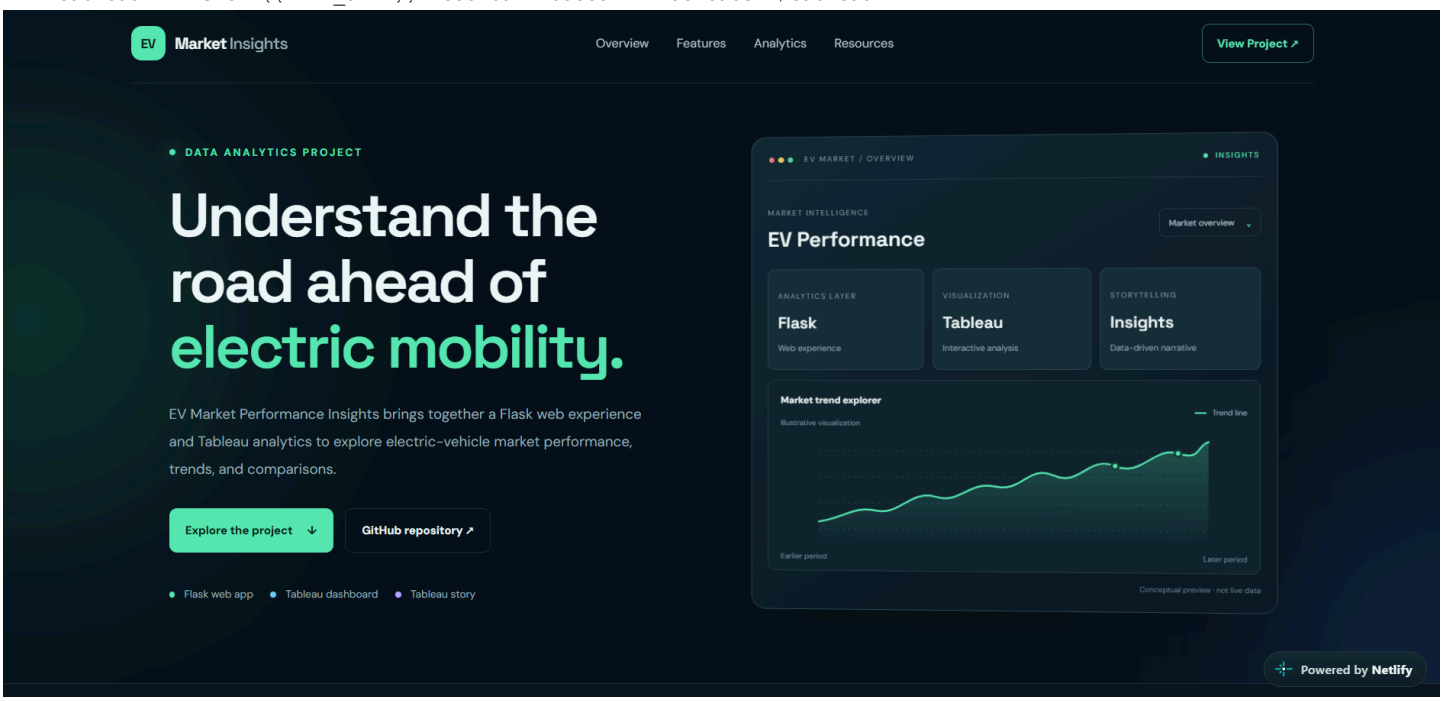


Figure 7: Home page running at 127.0.0.1:5000.

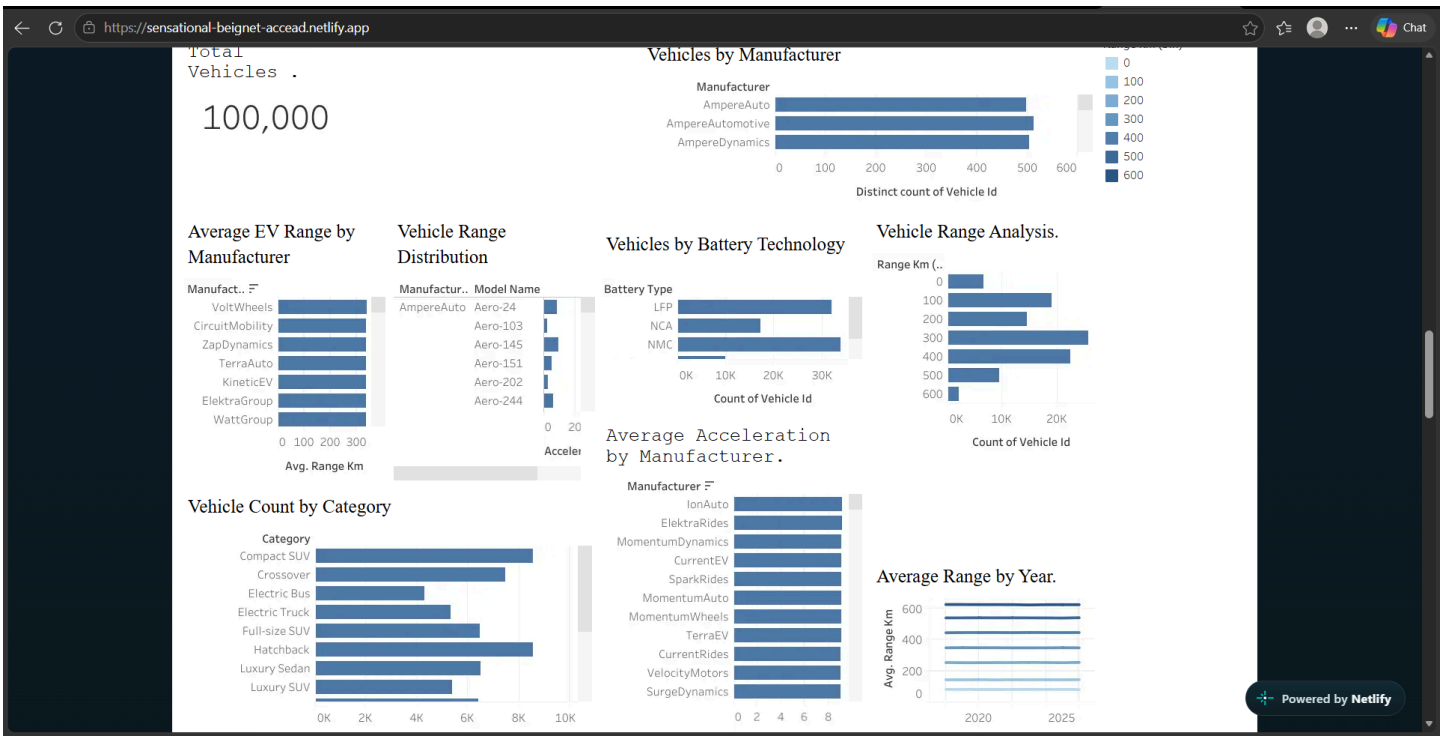


Figure 8: Dashboard page with the embedded Tableau dashboard and filters.

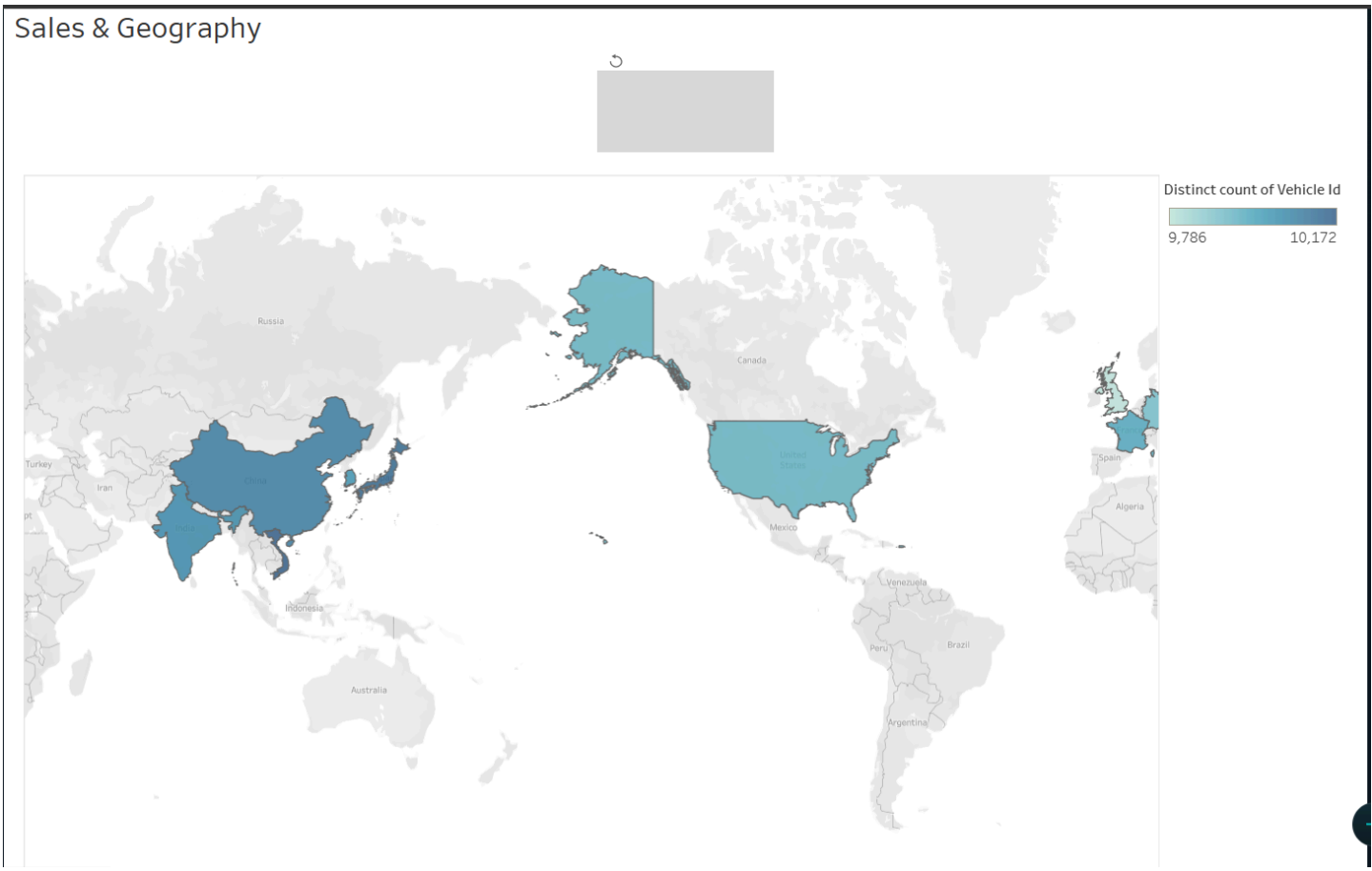


Figure 9: Story page with the six-scene Tableau story embedded.

10. Performance testing

Parameter	Value
Rows rendered	100,000
Columns	37
Data filters used	6
Calculated fields	6
Visualizations	12
Story scenes	6
Web pages (Flask)	4

To keep the dashboards responsive: charts that show 200 manufacturers are limited with a Top N filter, filters are shared across worksheets, and the scatter plot uses a single mark per model only where needed.

11. Key insights

- **Scooters lead volume:** two-wheeler scooters sell about 85.0M units, followed by sedans (78.3M) and compact SUVs (76.1M). Electric trucks, pickups and buses sell about 5 to 7M each.
- **Price follows battery size:** price correlates 0.84 with battery capacity but only 0.28 with range. Electric buses (average about $274k) and trucks (about $169k) are the most expensive.
- **Battery durability:** Solid-State averages 2,999 charge cycles and 7.36% degradation over five years; LFP follows with 2,498 cycles and 10.11%. NMC, Sodium-Ion and NCA are between 1,598 and 1,799 cycles with about 14% degradation.
- **No fast-charge or range gap:** every battery type averages about 21 minutes for DC fast charging and 320 to 325 km of range.
- **Balanced market:** the top five manufacturers (SurgeWorks, CurrentMobility, ElektraMobility, VelocityMobility, WattDynamics) sell 4.6M to 4.9M units each, and regions are evenly split.
- **Stable trend:** average price (about $65k) and range (about 324 km) are flat from 2018 to 2026.

Limitation: the dataset is synthetic, so summed sales (about 773 million units) and estimated revenue are relative measures, not real-world totals.

12. Conclusion and future scope

The project takes EV market data from a CSV file into a database, analyses it with SQL, visualizes it in Tableau and publishes the dashboard and story on a Flask website. The analysis shows which categories sell most, that battery size drives price, and that battery chemistry matters most for durability.

Future scope

- Add real-world data (registrations, charging stations) to replace synthetic fields.
- Add sales forecasting by category.
- Connect Tableau directly to a MySQL server with Tableau Desktop.
- Deploy the Flask app to a cloud platform.

13. Appendix

https://github.com/jainpragya2411-hub/EVM-project.git

GitHub repository:

https://github.com/jainpragya2411-hub/EVM-project.git

Dataset:

https://www.kaggle.com/datasets/tharishreddy22/electric-vehicle-market-classification-dataset

The inspected local files are ev_market_train.csv with 200,000 rows and ev_market_test.csv with 100,000 rows. Neither source CSV is included in the public repository. The project owner confirmed checking the dataset terms before publishing the Tableau extract; the license statement was not independently verified during this project review.

Public project and Tableau links

- GitHub Pages project website

https://comfy-cassata-cace86.netlify.app

- Tableau Public profile

https://public.tableau.com/app/profile/pragya.jain1129/vizzes

- Tableau Public Dashboard 1

https://public.tableau.com/views/Electric_Vehicle_Market_Insights_Sam_17931251104380/ElectricVehicleMarketPerformanceInsights?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

- Tableau Public Story 2

https://public.tableau.com/views/Electric_Vehicle_Market_Insights_Final_17931252349610/SalesGeography?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link